Lab08.

Ashley Osuna ; asosuna@ucsd.edu

```
#saved input data file into Project
#and assigned the data to fna.data
fna.data<-"WisconsinCancer.csv"

#input data and store as wisc.df
wisc.df<-read.csv(fna.data,row.names=1)
```

```
#ensure labels are set correctly
head(wisc.df,4)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38         122.80    1001.0
842517           M       20.57        17.77         132.90    1326.0
84300903         M       19.69        21.25         130.00    1203.0
84348301         M       11.42        20.38          77.58     386.1
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
84348301         0.14250          0.28390         0.2414             0.10520
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
84348301        0.2597                0.09744    0.4956     1.1560        3.445
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
84348301   27.23      0.009110        0.07458      0.05661           0.01867
         symmetry_se fractal_dimension_se radius_worst texture_worst
```

```
842302       0.03003            0.006193        25.38         17.33
842517       0.01389            0.003532        24.99         23.41
84300903     0.02250            0.004571        23.57         25.53
84348301     0.05963            0.009208        14.91         26.50
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
```

```r
#get rid of first column
wisc.data<-wisc.df[,-1]
```

```r
#creating vector with data from diagnosis
#data points from original dataset
diagnosis<-factor(wisc.df$diagnosis)
```

Q1. How many observations are in this dataset?

Answer Q1 : There are 569 observations

```r
nrow(wisc.df)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

Answer Q2 : There are 212 malignant diagnosis

```r
table(diagnosis)
```

```
diagnosis
  B   M
357 212
```

Q3. How many variables/features in the data are suffixed with _mean?

Answer Q3 : There are 10 of those variables

```r
sum(grepl("_mean",colnames(wisc.df)))
```

```
[1] 10
```

```r
#check column means and sd
colMeans(wisc.data)
```

```
            radius_mean            texture_mean          perimeter_mean
           1.412729e+01            1.928965e+01            9.196903e+01
              area_mean         smoothness_mean        compactness_mean
           6.548891e+02            9.636028e-02            1.043410e-01
         concavity_mean     concave.points_mean           symmetry_mean
           8.879932e-02            4.891915e-02            1.811619e-01
 fractal_dimension_mean               radius_se              texture_se
           6.279761e-02            4.051721e-01            1.216853e+00
           perimeter_se                 area_se           smoothness_se
           2.866059e+00            4.033708e+01            7.040979e-03
         compactness_se            concavity_se       concave.points_se
           2.547814e-02            3.189372e-02            1.179614e-02
            symmetry_se    fractal_dimension_se            radius_worst
           2.054230e-02            3.794904e-03            1.626919e+01
          texture_worst         perimeter_worst              area_worst
           2.567722e+01            1.072612e+02            8.805831e+02
       smoothness_worst       compactness_worst         concavity_worst
           1.323686e-01            2.542650e-01            2.721885e-01
   concave.points_worst          symmetry_worst fractal_dimension_worst
           1.146062e-01            2.900756e-01            8.394582e-02
```

```r
apply(wisc.data,2,sd)
```

```
            radius_mean            texture_mean          perimeter_mean
           3.524049e+00            4.301036e+00            2.429898e+01
              area_mean         smoothness_mean        compactness_mean
```

```
           3.519141e+02            1.406413e-02            5.281276e-02
         concavity_mean     concave.points_mean           symmetry_mean
           7.971981e-02            3.880284e-02            2.741428e-02
 fractal_dimension_mean               radius_se              texture_se
           7.060363e-03            2.773127e-01            5.516484e-01
           perimeter_se                 area_se           smoothness_se
           2.021855e+00            4.549101e+01            3.002518e-03
         compactness_se            concavity_se       concave.points_se
           1.790818e-02            3.018606e-02            6.170285e-03
            symmetry_se    fractal_dimension_se            radius_worst
           8.266372e-03            2.646071e-03            4.833242e+00
          texture_worst         perimeter_worst              area_worst
           6.146258e+00            3.360254e+01            5.693570e+02
       smoothness_worst       compactness_worst         concavity_worst
           2.283243e-02            1.573365e-01            2.086243e-01
   concave.points_worst          symmetry_worst fractal_dimension_worst
           6.573234e-02            6.186747e-02            1.806127e-02
```

```r
#Perform PCA on original data
#to ensure the input variables use different
#units of measurment and have significant
#different variables
wisc.pr<-prcomp(wisc.data,scale=TRUE)
```

```r
#visualize summary of outputs
summary(wisc.data)
```

```
  radius_mean      texture_mean   perimeter_mean     area_mean
 Min.   : 6.981   Min.   : 9.71   Min.   : 43.79   Min.   : 143.5
 1st Qu.:11.700   1st Qu.:16.17   1st Qu.: 75.17   1st Qu.: 420.3
 Median :13.370   Median :18.84   Median : 86.24   Median : 551.1
 Mean   :14.127   Mean   :19.29   Mean   : 91.97   Mean   : 654.9
 3rd Qu.:15.780   3rd Qu.:21.80   3rd Qu.:104.10   3rd Qu.: 782.7
 Max.   :28.110   Max.   :39.28   Max.   :188.50   Max.   :2501.0
 smoothness_mean    compactness_mean   concavity_mean    concave.points_mean
 Min.   :0.05263   Min.   :0.01938   Min.   :0.00000   Min.   :0.00000
 1st Qu.:0.08637   1st Qu.:0.06492   1st Qu.:0.02956   1st Qu.:0.02031
 Median :0.09587   Median :0.09263   Median :0.06154   Median :0.03350
 Mean   :0.09636   Mean   :0.10434   Mean   :0.08880   Mean   :0.04892
 3rd Qu.:0.10530   3rd Qu.:0.13040   3rd Qu.:0.13070   3rd Qu.:0.07400
 Max.   :0.16340   Max.   :0.34540   Max.   :0.42680   Max.   :0.20120
 symmetry_mean    fractal_dimension_mean   radius_se        texture_se
```

```
 Min.   :0.1060   Min.   :0.04996        Min.   :0.1115   Min.   :0.3602  
 1st Qu.:0.1619   1st Qu.:0.05770        1st Qu.:0.2324   1st Qu.:0.8339  
 Median :0.1792   Median :0.06154        Median :0.3242   Median :1.1080  
 Mean   :0.1812   Mean   :0.06280        Mean   :0.4052   Mean   :1.2169  
 3rd Qu.:0.1957   3rd Qu.:0.06612        3rd Qu.:0.4789   3rd Qu.:1.4740  
 Max.   :0.3040   Max.   :0.09744        Max.   :2.8730   Max.   :4.8850  
  perimeter_se       area_se        smoothness_se      compactness_se    
 Min.   : 0.757   Min.   :  6.802   Min.   :0.001713   Min.   :0.002252  
 1st Qu.: 1.606   1st Qu.: 17.850   1st Qu.:0.005169   1st Qu.:0.013080  
 Median : 2.287   Median : 24.530   Median :0.006380   Median :0.020450  
 Mean   : 2.866   Mean   : 40.337   Mean   :0.007041   Mean   :0.025478  
 3rd Qu.: 3.357   3rd Qu.: 45.190   3rd Qu.:0.008146   3rd Qu.:0.032450  
 Max.   :21.980   Max.   :542.200   Max.   :0.031130   Max.   :0.135400  
  concavity_se     concave.points_se   symmetry_se       fractal_dimension_se
 Min.   :0.00000   Min.   :0.000000   Min.   :0.007882   Min.   :0.0008948   
 1st Qu.:0.01509   1st Qu.:0.007638   1st Qu.:0.015160   1st Qu.:0.0022480   
 Median :0.02589   Median :0.010930   Median :0.018730   Median :0.0031870   
 Mean   :0.03189   Mean   :0.011796   Mean   :0.020542   Mean   :0.0037949   
 3rd Qu.:0.04205   3rd Qu.:0.014710   3rd Qu.:0.023480   3rd Qu.:0.0045580   
 Max.   :0.39600   Max.   :0.052790   Max.   :0.078950   Max.   :0.0298400   
  radius_worst   texture_worst   perimeter_worst    area_worst    
 Min.   : 7.93   Min.   :12.02   Min.   : 50.41   Min.   : 185.2  
 1st Qu.:13.01   1st Qu.:21.08   1st Qu.: 84.11   1st Qu.: 515.3  
 Median :14.97   Median :25.41   Median : 97.66   Median : 686.5  
 Mean   :16.27   Mean   :25.68   Mean   :107.26   Mean   : 880.6  
 3rd Qu.:18.79   3rd Qu.:29.72   3rd Qu.:125.40   3rd Qu.:1084.0  
 Max.   :36.04   Max.   :49.54   Max.   :251.20   Max.   :4254.0  
 smoothness_worst  compactness_worst concavity_worst  concave.points_worst
 Min.   :0.07117   Min.   :0.02729   Min.   :0.0000   Min.   :0.00000     
 1st Qu.:0.11660   1st Qu.:0.14720   1st Qu.:0.1145   1st Qu.:0.06493     
 Median :0.13130   Median :0.21190   Median :0.2267   Median :0.09993     
 Mean   :0.13237   Mean   :0.25427   Mean   :0.2722   Mean   :0.11461     
 3rd Qu.:0.14600   3rd Qu.:0.33910   3rd Qu.:0.3829   3rd Qu.:0.16140     
 Max.   :0.22260   Max.   :1.05800   Max.   :1.2520   Max.   :0.29100     
 symmetry_worst   fractal_dimension_worst
 Min.   :0.1565   Min.   :0.05504        
 1st Qu.:0.2504   1st Qu.:0.07146        
 Median :0.2822   Median :0.08004        
 Mean   :0.2901   Mean   :0.08395        
 3rd Qu.:0.3179   3rd Qu.:0.09208        
 Max.   :0.6638   Max.   :0.20750        
```

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)? **Answer Q4 : PC1 shows 44.27% of the original variance**

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data? **Answer Q5 : 3 PC's are required to describe for the minimum of 70% of variance.**

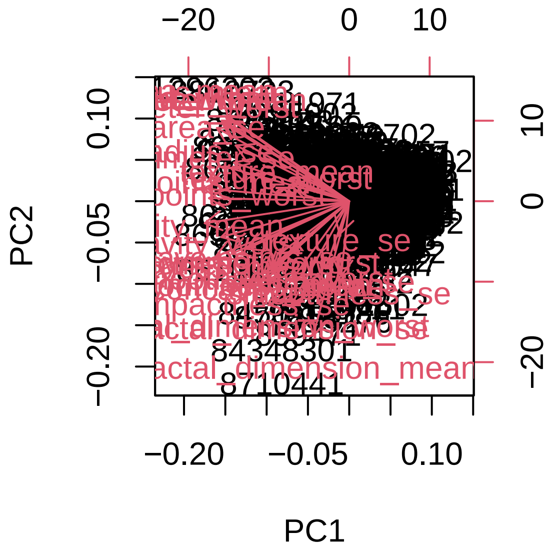
Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data? **Answer Q6 : We need 7 PC's to describe at least 90% of the variance based on PC7 reaching 0.91 (91%)**

```r
summary(wisc.pr)
```

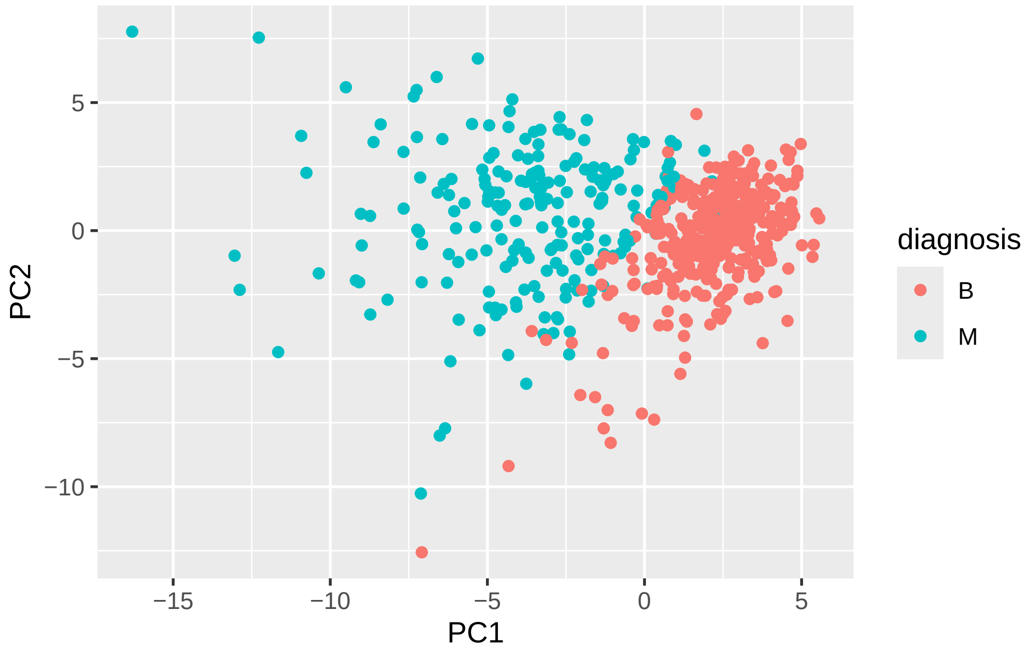
```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why? **Answer Q7 :Based on the biplot, the data points are not easily read in the plot. It is too crowded.**

```r
biplot(wisc.pr)
```

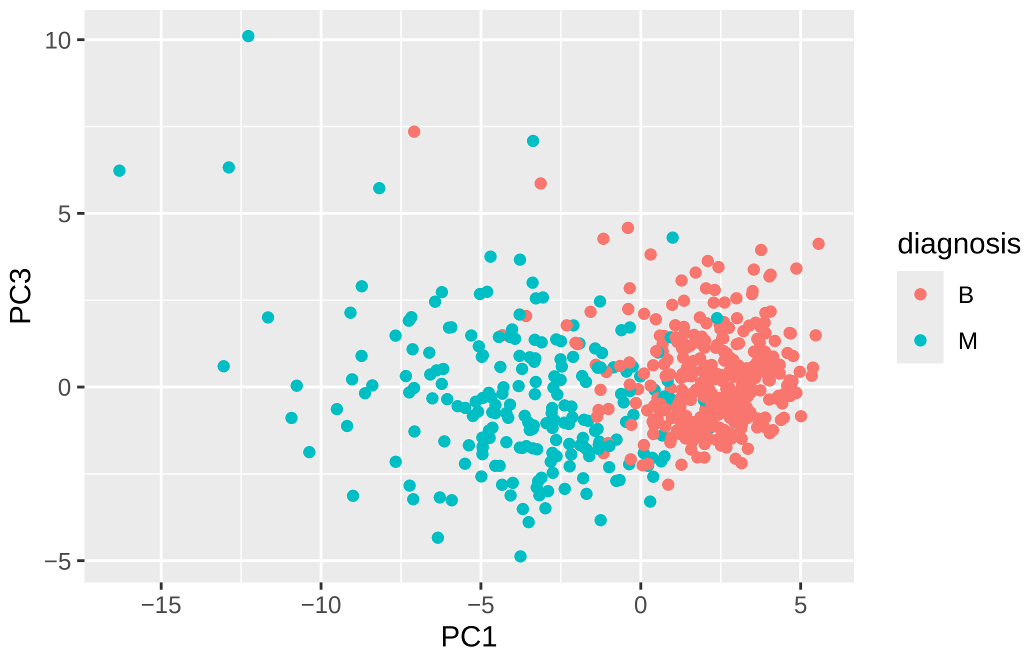



```r
#generate scatter plot by
#components one and two (cleaner graph)
library(ggplot2)



ggplot(wisc.pr$x)+
  aes(PC1,PC2,col=diagnosis)+
  geom_point()
```



Q8 : Generate a scatter plot for components 1 and 3. What do you notice about the plots?
Answer Q8 : PC2 proves to have more variance since it is easier to visualize the seperation from the two groups compared to the plot with PC1 and PC3

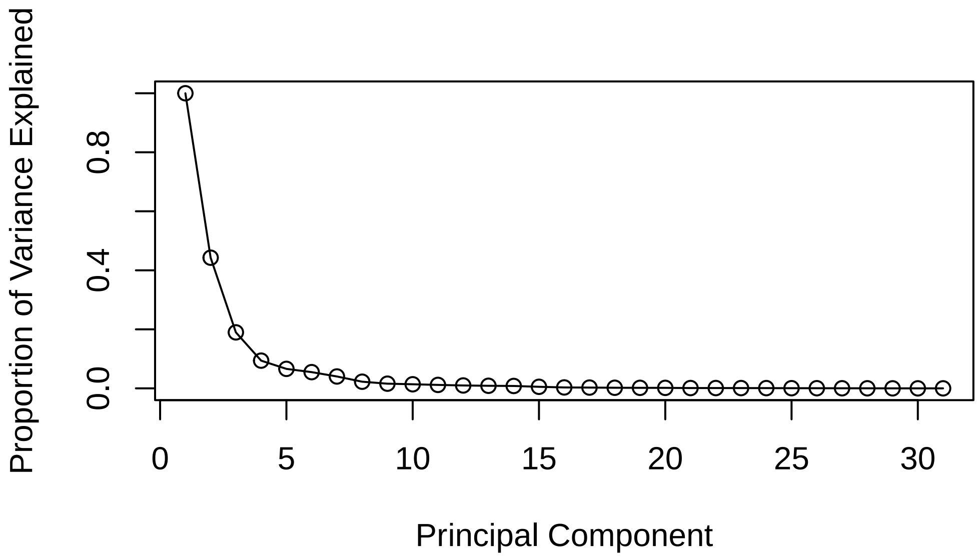
```
ggplot(wisc.pr$x)+
  aes(PC1,PC3,col=diagnosis)+
  geom_point()
```



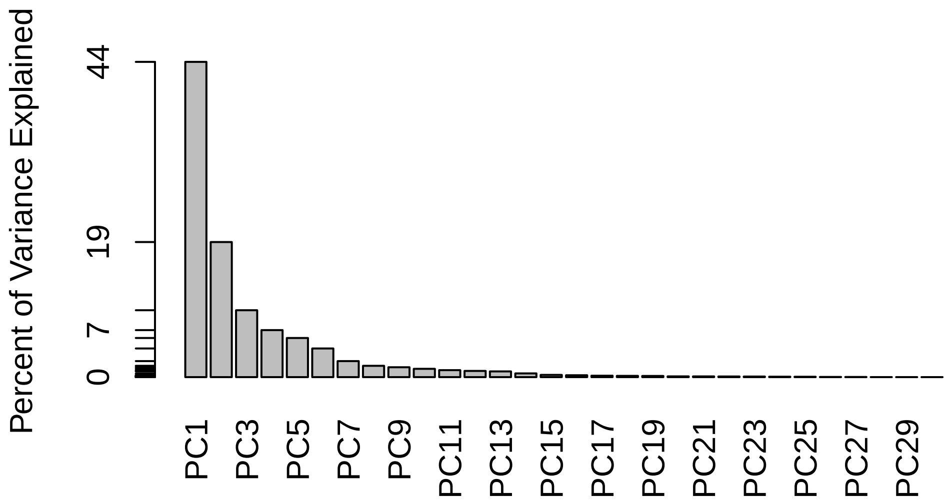

```
#calculate variance for each component
pr.var<-wisc.pr$sdev^2
#variance is sdev squared
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
pve<-pr.var/sum(pr.var)

#plot variance explained for
#each principal component
plot(c(1,pve),xlab="Principal Component",
     ylab="Proportion of Variance Explained",
     ylim=c(0,1),type="o")
```




```r
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab =
          "Percent of Variance Explained",
     names.arg=paste0("PC",1:length(pve)),
     las=2, axes = FALSE)
axis(2, at=pve,
     labels=round(pve,2)*100 )
```



Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

Answer Q9 : This feature contributes -0.2608538 to PC1. There are other features with much larger contributions (more negative), definitely

```
wisc.pr$rotation["concave.points_mean",1]
```

```
[1] -0.2608538
```

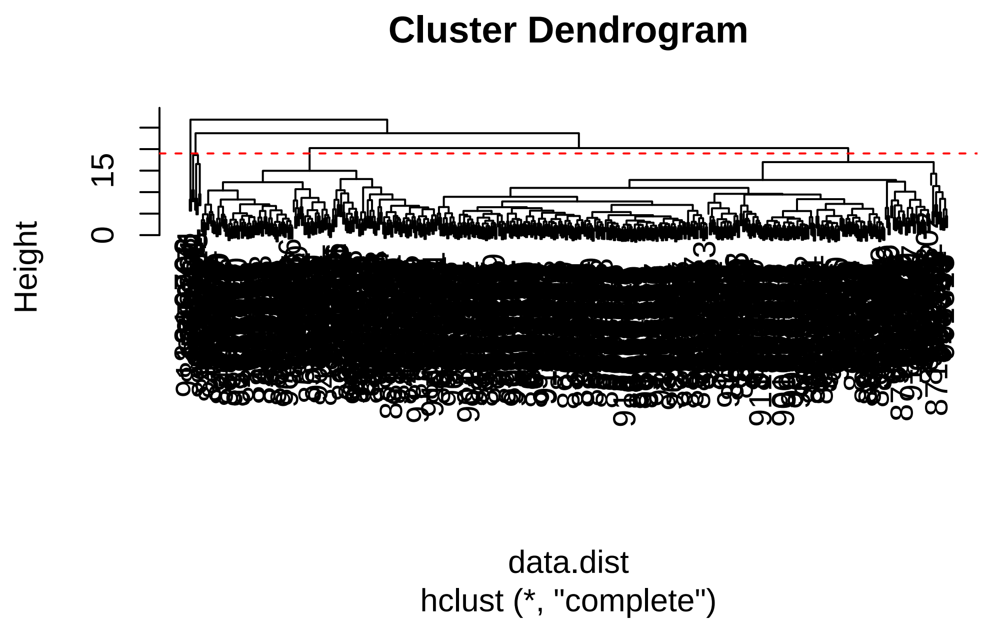
```
#scale wisc.data
data.scaled<-scale(wisc.data)
```

```
#calculate the distance between
#all pairs in new scaled dataset
data.dist<-dist(data.scaled)
```

```
#create hierarchical clutering model
wisc.hclust<-hclust(data.dist,method="complete")
```

Q10. optional

```r
plot(wisc.hclust)
abline(h=19,col="red",lty=2)
```




```r
#cut tree to have 4 clusters
wisc.hclust.clusters<-cutree(wisc.hclust,k=4)
```

```r
#compare clusters
table(wisc.hclust.clusters)
```

```
wisc.hclust.clusters
  1   2   3   4 
177   7 383   2 
```

Q11 - optional

Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning. **Answer Q12 :(code below) ward.D2 si the one I prefer since it had minimized variance within clusters to make evenly sized clusters. So there is a cleaner seperation within the data.**

```
wisc.hclust.single<-hclust(data.dist,
                           method="single")

wisc.hclust.complete<-hclust(data.dist,
                             method="complete")

wisc.hclust.average<-hclust(data.dist,
                            method="average")

wisc.hclust.wardt<-hclust(data.dist,
                          method="ward.D2")
```

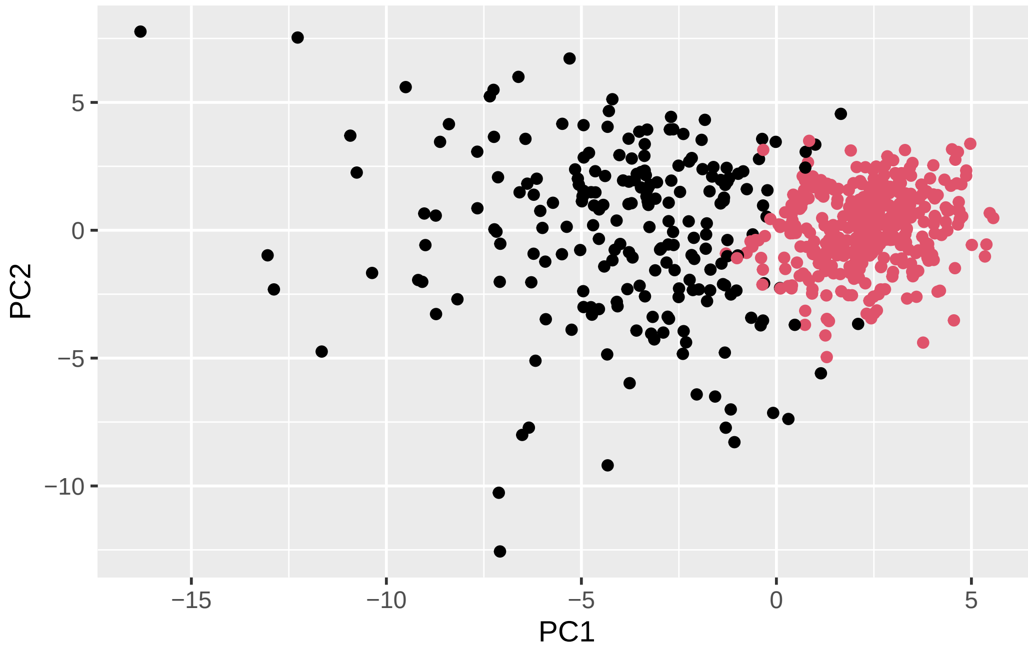
```
#assign wisc.pr.hclust with the ward.D2 method
#acknowledging the distance along PC1-PC7
wisc.pr.hclust<-hclust(dist(wisc.pr$x[,
                                      1:7]),
                       method="ward.D2")
grps<-cutree(wisc.pr.hclust,
             k=2)
table(grps)
```

```
grps
  1   2
216 353
```

```
table(grps,diagnosis)
```

```
    diagnosis
grps   B   M
   1  28 188
   2 329  24
```

```
ggplot(wisc.pr$x)+
  aes(PC1,PC2)+
  geom_point(col=grps)
```



optional section below

```
#confirming that previously used
#distance along the first 7 PCs for clustering
wisc.pr.hclust<-hclust(dist(wisc.pr$x[
  ,1:7]),method="ward.D2")
```

```
wisc.pr.hclust.clusters<-cutree(
  wisc.pr.hclust,k=2)
```

Q13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses? **Answer Q13 : Cluster 1 has 188 malignant and only 28 benign, while cluster 2 has 329 benign with 24 malignant. Therefore, most samples had been grouped.**

```
table(wisc.pr.hclust.clusters,diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

```r
table(wisc.pr.hclust.clusters,diagnosis)
```

```
                      diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

Answer 14 : The original hierarchical clustering is worse in comparison as it splits 4 clusters and is much difficult to visualize for interpretation. The PCA-model is much cleaner to distinguish the data withi the clusters.

```r
table(wisc.hclust.clusters,diagnosis)
```

```
                   diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

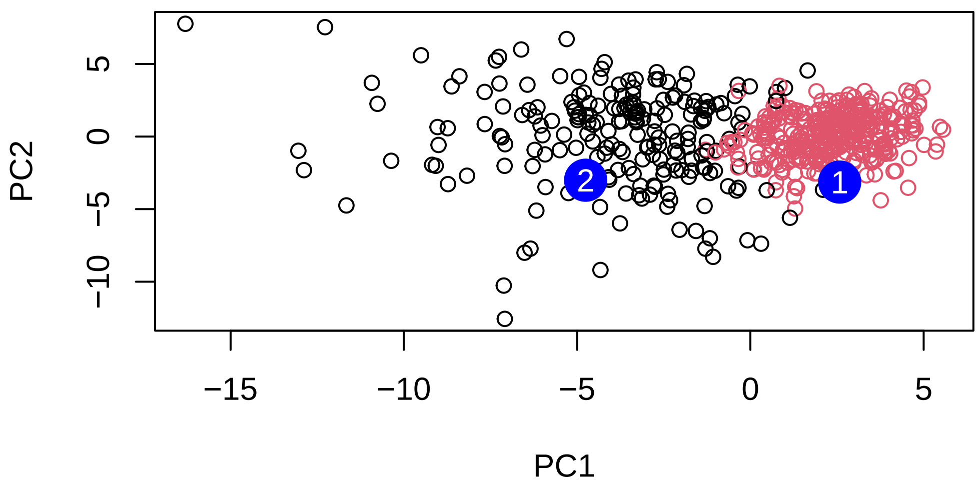
Q15 optional

```r
#use predict() function for PCA space
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
```

```
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```r
#plot new data
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?
Answer Q16 : Patient 2 demonstrates to be more prioritized for the follow up based on the black cluster (group 1) which relates to malignant diagnoses. Patient 1, however, is represented as the pink cluster which is benign.